Assessing the Performance of Large Language Models on the Foreign Medical Graduate Examination (FMGE): Insights from GPT-4 Turbo, Gemini Advanced, and LLaMA 3.1 (70B)

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Abstract*—This study evaluates three advanced Large Language Models (LLMs)—GPT-4 Turbo, Gemini Advanced, and Meta's LLaMA 3.1 (70B)—for accurately answering multiple-choice questions from the Foreign Medical Graduate Examination (FMGE). Using a curated set of 427 text-based questions from recent exams, the investigation assessed each model's overall accuracy, consistency, and error distribution, with statistical validation via McNemar's test. GPT-4 Turbo achieved roughly 93% accuracy, while Gemini Advanced and LLaMA 3.1 (70B) both approximated 87%, with no significant difference between them. High agreement among models indicates consistent decision-making and stable interpretative patterns. These findings underscore the potential of LLMs as complementary tools for exam preparation, particularly in resource-limited settings. Moreover, the results support the viability of open-source models—exemplified by Meta's LLaMA 3.1 (70B)—in terms of cost-effectiveness and adaptability. Future research will explore diverse question formats and the integration of these models into clinical decision support systems to further enhance their role in modern medical education.

***Keywords*— Large Language Models (LLMs), Foreign Medical Graduate Examination (FMGE), Medical Licensing Examinations, GPT-4 Turbo, LLaMA 3.1 (70B), Gemini Advanced, AI in Medical Education.**

I. Introduction

Introduced in 2002, the Foreign Medical Graduate Examination (FMGE) is a screening test for Indian citizens and Overseas Citizens of India (OCIs) who hold primary medical qualifications from recognized foreign institutions. Passing the FMGE is a prerequisite for provisional or permanent registration with the National Medical Commission (NMC) or any State Medical Council [1]. The FMGE aims to ensure that these graduates possess the necessary knowledge and skills to provide quality healthcare services in India [2]

Administered biannually, typically in June and December, the computer-based FMGE comprises 300 multiple-choice questions (MCQs) divided equally into two sections of 150 questions each, with no negative marking. The syllabus mirrors that of the MBBS program in India, covering pre-clinical, para-clinical, and clinical disciplines, while official question papers and answer keys are not released [2].

The FMGE presents significant challenges due to its high difficulty level, with many candidates facing resource constraints from the unavailability of official question papers and limited access to high-quality study materials. This preliminary study investigates the application of artificial intelligence (AI) tools to enhance FMGE preparation strategies.

Artificial Intelligence denotes the development of systems that perform tasks typically requiring human cognition such as learning, problem-solving, pattern recognition, and decision-making [3]. Among AI technologies, Large Language Models (LLMs), a subset of generative AI are designed to process and generate human-like text using deep learning and natural language processing techniques. This renders LLMs particularly useful for examination preparation, as they can produce detailed explanations of academic and technical concepts [4], [5].

LLMs have demonstrated advanced capabilities across various national and international medical examinations. Previous studies have primarily examined the performance of OpenAI's GPT models (e.g., GPT-3.5 and GPT-4) and other proprietary models concerning various international medical exams, leaving their evaluation in answering FMGE questions underexplored. Moreover, open-source LLMs, which are advantageous for resource-constrained systems, have received limited research attention. This study seeks to address this gap by evaluating the performance of three advanced LLMs – GPT-4 Turbo, Meta LLaMA 3.1 (70B),

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and Google Gemini Advanced in answering FMGE MCQs. This study benchmarks the performance of LLMs in answering FMGE MCQs, using accuracy as a proxy measure of the reliability of LLM-generated answer keys. The evaluation aims to determine if students can depend on these responses for FMGE preparation.

Emphasis is placed on open-source LLMs due to their cost-effectiveness and accessibility in the Indian context, addressing the need for reliable AI-driven support for students with limited access to traditional resources and private coaching.

The study's significance is underscored by a Harvard Medicine Magazine article, which described generative AI as the most consequential development in medical education since the advent of the Internet [6]. Harvard University has been a pioneer in integrating generative AI into its medical curriculum, recognizing its transformative impact.

For this analysis, a dataset was compiled from previous FMGE question papers. Out of 600 questions, 427 were selected after excluding image-based questions (given Meta's LLaMA limitations) and those that were ambiguous, incomplete, or insufficiently detailed.

II. Related works

Large language models are increasingly applied in healthcare and medical education [4], [7], [8]. Since ChatGPT's introduction in November 2022, continuous advancements have enhanced LLMs' abilities to understand inputs and generate human-like responses [4], [5], [9]. These models have been integrated into various medical fields including oncology, pathology, and radiology to improve diagnostics, support clinical decision-making, and enhance patient care. In medical education, LLMs show promise by generating logical justifications with their answers, serving as supplementary learning tools, offering personalized tutoring, and creating relevant clinical scenarios [4], [5], [7], [9], [10].

However, LLMs may also produce inaccurate or misleading information with unwarranted confidence. Their limited capacity for critical reasoning and tendency to provide excessively verbose responses can be problematic [4] ,[7], [9]. Furthermore, inherent biases in their training data and methodologies may be reinforced, rendering them unreliable as the sole resource for exam preparation [9].

Licensing and advanced medical examinations offer valuable benchmarks for assessing LLM capabilities, as they evaluate a broad spectrum of medical knowledge, practical skills, and clinical reasoning [11], [12]. In the early stages of public attention, USMLE questions were used to gauge LLM potential and limitations [13]. For example, ChatGPT-4 achieved scores at or near the USMLE passing threshold without any domain-specific training [14]. According to [10], GPT-4 reached nearly 100% accuracy on sample National Board of Medical Examiners (NBME) clinical exam questions, significantly outperforming models such as GPT-3.5, Claude, and Google Bard. Additionally, ChatGPT has demonstrated deductive reasoning and a coherent chain-of-thought process, which can be advantageous for medical training [11], [12]. A study [15] reported that GPT-4 attained a 76.3% accuracy rate on the UK Medical Licensing Assessment (UKMLA), surpassing the average candidate passing rate of 70% and indicating its qualifying potential.

The performance of LLMs on medical licensing examinations across Japan, India, China, Peru, and Poland offers insights into the influence of cultural context and regional curricula on their outcomes. For instance, a study evaluating GPT-4 on the Chinese National Medical Licensing Examination reported a score of 61.6% (weighted F1 score of 0.617), notably lower than the human average of approximately 71.6% [16]. Similarly, on the Japanese National Medical Licensing Examination, ChatGPT-4 achieved 81.5%, surpassing the passing threshold, while GPT-3.5 scored only 42.8% [17].

In India's postgraduate medical entrance examination, GPT-4 attained 64.5% (correctly answering 129 out of 200 questions), placing it within a national rank range of 20,000 to 30,000 and rendering it non-competitive for preferred postgraduate specializations [18]. Conversely, in the Peruvian National Medical Licensing Examination, GPT-4 demonstrated expert-level performance with an accuracy of 86%, outperforming 90% of human candidates, compared to GPT-3.5's 77% [11]. Furthermore, evaluations on the Polish Medical Final Examination (MFE) reveal regional performance disparities: GPT-4 achieved an average accuracy of 79.7% on both Polish and English versions, successfully passing all variations, while GPT-3.5 recorded averages of 54.8% in Polish and 60.3% in English—failing all but two of the Polish versions, despite passing all English versions, given a minimum passing threshold of 56% [12].

These findings reveal significant variation in LLM performance across national medical examinations, highlighting limitations in universal applicability and standardized benchmarking. Variations in exam difficulty, content diversity, and language-specific challenges contribute to these disparities. Consequently, region-specific training in non-English languages is warranted to improve international applicability. Integrating translation, local terminologies, and region-specific medical challenges into the training process is essential to enhance model accuracy [9].

The most common metric for evaluating LLMs is the percentage of correct responses [4], [5], [11]. However, definitions of accuracy vary: some studies deem a response successful if it matches the official answer key, while others assess qualitative factors such as reasoning depth, coherence, and clarity of explanations [4], [11], [13]. A few studies have employed statistical tests, like one-way ANOVA, to determine whether one LLM significantly outperforms others [10].

Additionally, exam format influences LLM performance evaluation. Several studies used multiple-choice questions, while a subset employed open-ended and scenario-based questions. Although multiple-choice items are limited in scope, open-ended formats assess broader understanding, analytical skills, and the application of medical knowledge [9 -10]. Moreover, performance in non-English language contexts was significantly lower, with notably suboptimal

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outcomes in traditional Chinese medicine and medicinal Chinese pharmacy [19].

Identified Research Gaps for this Study

While previous research has evaluated LLM performance on exams like the USMLE, UKMLA, and other Indian national examinations, studies on the FMGE remain limited. To date, only one study has examined LLM performance on the FMGE in the context of Ophthalmology [20], underscoring the need to assess LLMs across the full spectrum of FMGE subjects.

Moreover, most benchmarking efforts have focused on proprietary models, with minimal comparison to open-source alternatives such as Meta's LLaMA. Although open-source models offer advantages in transparency, customization, and cost, their effectiveness in high-stakes medical assessments is underexplored. A comparative evaluation of open-source and proprietary LLMs is therefore essential.

Our study aims to bridge these research gaps by evaluating LLM performance on the FMGE. Given the rapid evolution of AI technology and the expanding capabilities of LLMs, such studies are crucial. These advancements have significant implications for delivering high-quality patient care, highlighting the need for ongoing assessments to better understand the strengths and limitations of these models [9].

III. METHODOLOGY

This study employs a comparative evaluation of three LLMs in answering FMGE exam MCQs, utilizing an experimental approach to assess both open-source and proprietary models. The evaluated models include Meta LLaMA 3.1 (70B) (open-source) and two proprietary models, ChatGPT-4 Turbo and Gemini Advanced. The methodology adopts a cross-sectional design with standardized data selection, structured evaluation, and statistical analysis to ensure a robust and unbiased comparison.

A. Study Design

This study was designed to ensure methodological rigor, fairness, and minimized bias in model responses. A standardized comparative framework was implemented by subjecting all three models to an identical set of FMGE MCQs under controlled conditions. Each MCQ was presented as text followed by four answer options. Prompts(refer Table I) were manually entered in batches of ten questions, with a new chat session initiated after each batch to prevent memory bias. The tests were conducted once per model by different authors at their respective workplaces, and all responses were documented. No follow-up questions or requests for reasoning were made. The LLM-generated answer keys were then compared with expert-validated keys. It is unlikely that the MCQs were part of the LLMs' training data, as their performance before and after the training cutoff date has exhibited no significant variation.

B. Data Selection

The FMGE examination authority does not release official questions or answer keys.

TABLE I: SAMPLE PROMPT USED IN THIS STUDY FOR EVALUATING LARGE LANGUAGE MODELS ON FMGE MCQS

You are an advanced large language model acting as an expert medical professional tasked with providing correct answer keys for multiple-choice questions (MCQs) derived from the FMGE (Foreign Medical Graduate Examination) in India. Your objective is to evaluate each question carefully and supply only the correct answer.

For each MCQ, follow these steps:

I. **Repeat the Question**: Restate the question for clarity and context.

II. **List Options with Labels**: Clearly display each answer choice, labelling them as (A), (B), (C), or (D).

III. **Use Chain of Thought Internally**: Apply the chain of thought reasoning process internally to arrive at the correct answer. Do not display the reasoning or explanation to maintain brevity.

IV. **Provide the Correct Answer**: State only the correct option and answer at the end, using the format:

"Correct Answer: [Option] [Answer]"

Therefore, we obtained exam papers for the January 2023 and July 2024 FMGE exams from publicly available sources provided by a coaching institution [21]. The MCQs recall-based questions reported by examinees were corrected using [22] and verified by medical experts for accuracy. Image-based questions were excluded, resulting in a dataset limited to text-based MCQs. We designate the January 2023 exam as D1 and the July 2024 exam as D2.

The final dataset comprises 427 questions (215 from D1 and 212 from D2), covering a broad range of medical subjects. The MCQs were organized by subject (e.g., anatomy, general medicine) and formatted in a spreadsheet for processing by the LLMs, with each question featuring one correct answer and three distractors.

IV. EXPERIMENTAL RESULTS

This investigation evaluated LLM performance using key metrics, including accuracy, statistical tests, consistency, agreement rates, and prediction differences. The analysis considered overall accuracy, performance across dataset subsets (D1 and D2), response consistency, error distribution, and comparisons with existing research.

A. Performance based on Overall Accuracy

The evaluated LLMs demonstrated high overall accuracy across all datasets. A 95% confidence interval (CI) was employed to ensure statistical robustness, with Table II presenting the accuracy and corresponding CIs for each model.

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TABLE II: MODEL ACCURACY AND CONFIDENCE INTERVALS FOR FMGE DATASETS

Model	Average Accuracy (%)	95% Confidence Interval
GPT-4 Turbo	92.98	[90.56%, 95.40%]
Gemini Advanced	87.36	[84.21%, 90.51%]
LLaMA 3.1 (70B)	87.12	[83.94%, 90.30%]

Notably, GPT-4 Turbo significantly outperformed the other models, while Gemini Advanced and LLaMA 3.1(70B) exhibited statistically indistinguishable performance.

B. Performance based on Dataset Subsets

The study evaluated GPT-4 Turbo, LLaMA 3.1 (70B), and Gemini Advanced across multiple datasets to assess their effectiveness in answering medical licensing exam questions. GPT-4 Turbo consistently achieved superior accuracy, while LLaMA 3.1 (70B) and Gemini Advanced displayed comparable, slightly lower performance levels. Fig. 1 illustrates these accuracy trends, and the variations observed across dataset subsets., highlighting variations across dataset subsets.

C. Performance based on Consistency in Responses

High agreement rates among models indicate consistent decision-making, reflecting stable interpretative patterns and reliable handling of similar datasets over time. As shown in Fig. 2, the GPT, LLaMA, and Gemini models exhibit a high degree of consistency in their responses across different time periods, with variations of less than 0.5% in most cases.

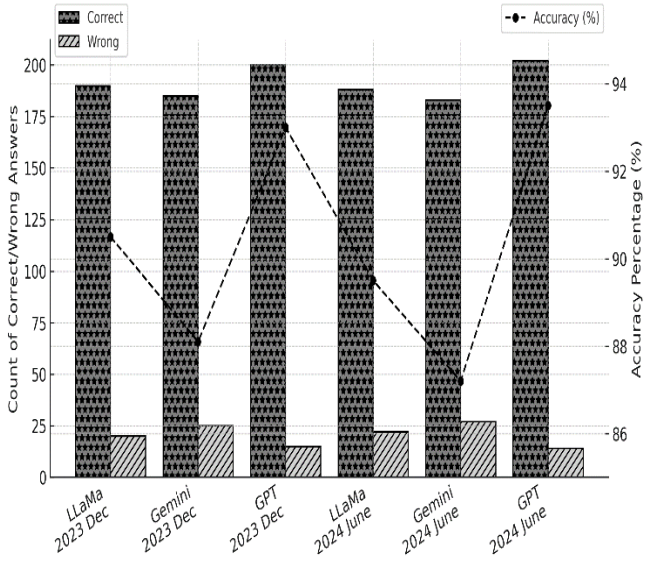


Fig 1: Accuracy Trends across dataset subsets

This stability suggests that despite potential updates or modifications, the models' responses have remained largely uniform. Consistently high agreement rates—exceeding 87% across all comparisons—demonstrate strong alignment in model interpretations, while minor fluctuations between individual model pairs do not indicate significant divergence. Uniform trends across comparisons (LLaMA-Gemini, LLaMA-GPT, and Gemini-GPT) further reinforce the robustness of these models in maintaining consistent output behaviour.

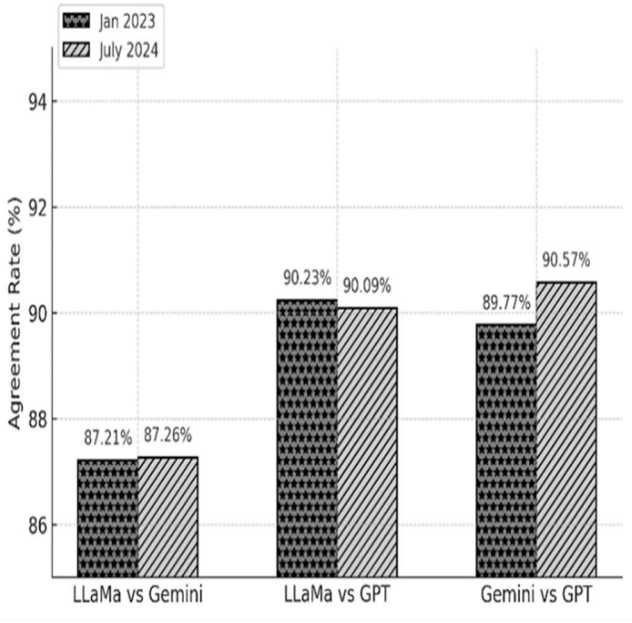


Fig. 2: Agreement Rates across Datasets

D. Performance based on Error Distribution and Statistical Significance

We applied McNemar's test on paired categorical data to determine if the models exhibit significantly different error patterns on the same dataset. This test uses the chi-square distribution to compute a p-value, with values below 0.05 indicating statistical significance. Table III shows the pairwise comparisons, where the chi-square statistic measures the magnitude of performance differences, and the p-value indicates the probability that these differences occurred by chance. Key findings are summarized.

2023 Results: No statistically significant differences were found between any of the three models (LLaMA, Gemini, and GPT-4). This means they performed similarly on the 2023 FMGE questions.

2024 Results: A statistically significant difference was observed between LLaMA and GPT-4. The data suggests one model likely outperformed the other. There was no significant difference between LLaMA and Gemini, or Gemini and GPT-4.

Combined Data: A statistically significant difference was found between LLaMA and GPT-4, with GPT-4 showing superior performance. No significant difference was found between LLaMA and Gemini, or Gemini and GPT-4.

The combined results reinforce that GPT-4 consistently outperformed LLaMA across both years, whereas LLaMA and Gemini, as well as Gemini and GPT-4, showed comparable performance.

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TABLE III: MCNEMAR'S TEST RESULTS FOR PAIRWISE MODEL COMPARISONS

Comparison	Chi-Square	P-Value
LLaMA vs Gemini (2023)	27	1
LLaMA vs GPT-4 (2023)	17	0.174171147
Gemini vs GPT-4 (2023)	17	0.135156451
LLaMA vs Gemini (2024)	26	0.891923151
LLaMA vs GPT-4 (2024)	14	0.043558522
Gemini vs GPT-4 (2024)	14	0.080690468
LLaMA vs Gemini (Combined)	0.5	1
LLaMA vs GPT-4 (Combined)	0.0217	0.0012
Gemini vs GPT-4 (Combined)	0.0208	0.8937

E. Coarse-Grained Analysis of Large Language Model Performance Across Medical Specialties

This analysis evaluated three large language models (LLMs) across 19 subjects of the MBBS curriculum in India. Due to the limited datasets used for each specialty, these findings should be interpreted with caution; a larger dataset combined with a more detailed qualitative analysis could yield deeper insights.

Overall GPT-4 made fewer errors across the medical specialties. The LLMs demonstrated high accuracy in fact-based disciplines, including anatomy, physiology, biochemistry, pathology, microbiology, anesthesia, psychiatry, and dermatology. In contrast, Preventive and Social Medicine (PSM) posed challenges: among 60 questions, LLaMA answered 16 incorrectly, Gemini answered 11 incorrectly, and GPT-4 answered only 5 incorrectly. This discrepancy likely reflects the broad scope of PSM—encompassing public health measures, epidemiology, and region-specific policies—as well as a lack of localized training on Indian guidelines.

In clinical specialties such as pediatrics, obstetrics and gynecology, medicine, and ENT, moderate error rates were observed, possibly due to the complex, case-based reasoning required in these fields. Among these, Gemini performed slightly worse than both GPT-4 and LLaMA. These issues may stem from limited domain-specific training as well as the inherent challenges that LLMs face in nuanced, multi-step clinical reasoning.

Furthermore, the similarity among multiple-choice options may have led to additional errors. Overall, although LLMs excel in well-documented, factual subjects, their performance declines in areas that require local expertise and complex reasoning. Therefore, it is inadvisable to rely solely on these models for critical medical decisions without cross-verification by subject matter experts.

F. *Comparison with Existing Approaches*

A comparison with similar studies evaluating LLMs on various medical licensing exams is shown in Table IV.

TABLE IV: COMPARISON WITH EXISTING APPROACHES

Exam Evaluated/[Study]	Accuracy (%)
Foreign Medical Graduate Examination (FMGE)/ This study	GPT-4 Turbo: 93.44 Gemini Advanced: 87.36 LLaMA3.1 (70B): 87.12
National Eligibility cum Entrance Test for Postgraduate (NEET PG) / [7]	GPT-4: 64.5%
University of Antwerp Medical Licensing Exam / [23]	GPT-4:76% ChatGPT-3.5: 67%.
United Kingdom Medical Licensing Assessment (UKMLA)/ [15]	GPT-4:76.3%
Saudi Medical Licensing Exam (SMLE) / [24]	GPT-4: 88.6%
United States Medical Licensing Exam (USMLE) / [9]	GPT-4: 84.6% GPT 3.5 56.9%
Chinese National Medical Licensing Examination (NMLE) / [16]	ChatGPT-4: 61.6%
Peruvian National Licensing Medical Examination (ENAM) / [11]	GPT-4:86% GPT-3.5: 77%
National Medical Licensure Examination (NMLE) (Japan) / [17]	GPT-4:81.5% GPT-3.5: 42.8%
Japanese National Medical Licensing Examination (JMLE) / [25]	GPT-4: 79.9% GPT-3.5: 50.8%

V. LIMITATONS AND FUTURE WORK

The study's scope was constrained by using a dataset consisting of text-based MCQs, which do not fully represent the complexity of the medical exams, namely the inclusion of image-based, scenario-driven, and open-ended question formats. Additionally, the evaluation primarily focused on accuracy metrics and did not examine the models' reasoning processes or their ability to adapt to dynamic clinical scenarios.

Future research shall expand the dataset to include a broader range of question formats and complexities, thereby providing a more comprehensive assessment of LLM performance. Further studies should focus on enhancing the models' reasoning capabilities and integrating LLMs into clinical decision support systems. Optimizing open-source models, such as Meta LLaMA, for region-specific curricula and diverse medical contexts may broaden their practical applications in medical education and clinical practice. Furthermore, because the models did not have internet access during the study, future evaluations should investigate the impact of internet-enabled models on performance, particularly about improvements in real-time information retrieval and accuracy.

VI. CONCLUSION

This study conducted a comparative evaluation of three advanced LLMs —GPT-4 Turbo, Meta LLaMA 3.1(70B), and Google Gemini Advanced—using FMGE multiple-choice questions as a benchmark. The analysis revealed that GPT-4 Turbo consistently outperformed its counterparts; however, the performance of Meta LLaMA 3.1(70B) was also notable, particularly given its open-source nature. Open-source models offer significant advantages in terms of

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Acessibility, cost-effectiveness, and customizability, rendering them especially attractive for resource-constrained settings. Overall, these findings underscore the potential of LLMs as complementary educational tools in medical training when used in conjunction with traditional learning methods.

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